Memo_3_index.qmd

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Research or Analytical Question:
How do communication apprehension scores differ between public transit users and non-users?

Answer, Response, + Summary of Results

This memorandum uses `merged_data.csv` to compare communication apprehension scores between public transit users and non-users.

Communication apprehension was measured using 12 Likert-scale items: `Q1`–`Q6` and `Q13`–`Q18`. Responses were coded from 1 for **Strongly disagree** to 5 for **Strongly agree**. Positively worded items (`Q2`, `Q4`, `Q6`, `Q14`, `Q16`, and `Q17`) were reverse-coded so that higher total scores consistently represented greater communication apprehension.

Participants who selected **Never** for `Q26` were coded as non-users. Participants selecting any other public-transit-use category were coded as users.

Table 1: Summary table of public transit users and non-users

transit_group	count	mean	std	median
Non-user	50	33.72	13.33	33.5
User	191	27.68	10.51	25.0

Public transit users had a mean communication apprehension score of **27.68** (`n = 191`), whereas non-users had a mean score of **33.72** (`n = 50`). Non-users therefore scored approximately **6.04 points higher**, on average, than transit users.

A visual comparison of the two groups indicates that communication apprehension scores tend to be higher among public transit non-users than among users.

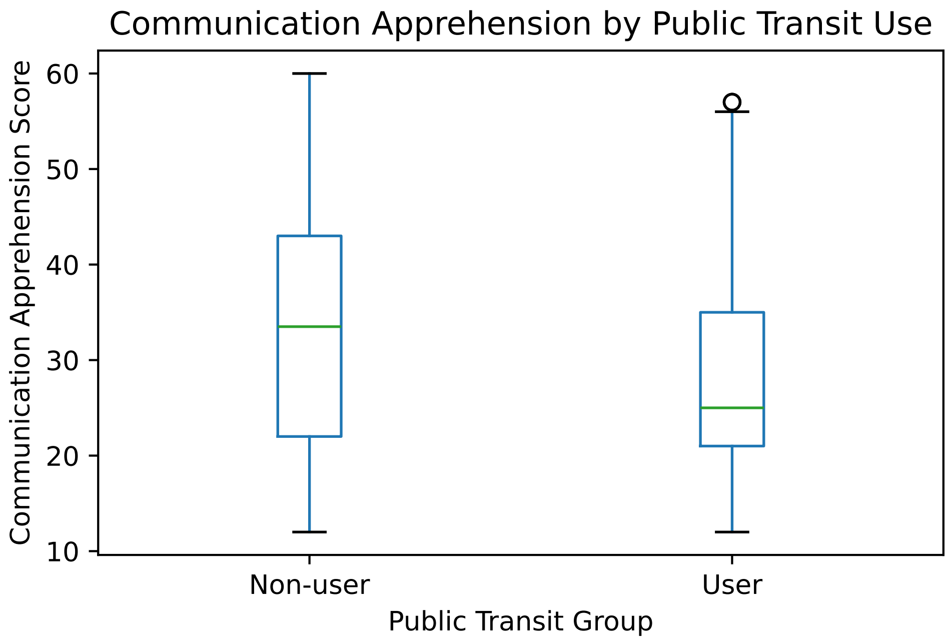

Figure 1: Communication apprehension scores among public transit users and non-users.

```
t = -2.971
df = 65.79
p = 0.0041
```

A Welch independent-samples t-test found a statistically significant difference between the groups, $t(65.79) = -2.97$, $p = .004$. Based on these data, public transit users reported lower communication apprehension scores than non-users.

This analysis identifies a difference between the two groups, but it does not establish that communication apprehension causes people to use or avoid public transportation.

What questions or uncertainties remain?

Do demographic characteristics, access to a private vehicle, or the availability of public transportation help explain the difference between users and non-users?

Would the result remain similar after controlling for age, employment status, student status, or country of residence?

Does the frequency of public transit use differ across levels of communication apprehension among participants who already use public transportation?